Supply Radar — Detailed Solution Report

Use Case 6 · Supplier & Sourcing Intelligence for Vaighai · predictions for FY2027 FQ2 · opportunity view FY2026

1. What the solution does

Supply Radar turns three data sources into a single supplier-side decision tool. It joins the MIR field estimates (market view), our ERP purchase records (what we actually bought), and regional rainfall, at the grain of one mill per region per quarter. It then answers four business questions with models, and writes the results up in plain English on a live dashboard with a weekly sourcing brief.

Pipeline: Extract (3 sources) to Bronze, Transform and clean to Silver, hard data-quality gates, load to the SQL warehouse, run the models to Gold, an AI agent writes the brief, and a dashboard presents it. One command, fully local, with a one-to-one path to Azure.

2. The five components (models)

1. Decline-risk classifier. Logistic regression on 25 features (recent volume, momentum, our share, seasonality, weather), probability-calibrated. Predicts the chance a mill's dispatch falls more than 50% below its own recent average next quarter. Output: a risk percentage and a Low / Moderate / Critical band.

2. Volume forecaster. Picks the most accurate of four methods each run — ridge regression, seasonal-naive, a blend, and Croston-TSB (for on-and-off mills) — and reports next-quarter tonnes with a P10 to P90 range.

3. Opportunity scorer. A transparent formula: mill size x how much room is left (headroom) x recent growth. Output: a ranked shortlist of where to buy more, with untapped tonnes.

4. Concentration tracker. HHI and top-5 supplier share per year — how dependent we are on a few mills.

5. Weather signal + AI agent. Rainfall (this quarter, lagged, cumulative monsoon, plus a live 16-day forecast) feeds the models; a language-model agent turns the final numbers into a plain-English weekly brief and on-screen explanations.

3. Model performance (honestly measured)

Validated by rolling-origin cross-validation over 6 quarters (train on the past, test the next quarter), with no future data leaking into training.

Model	What it measures	Result
Decline risk	Accuracy (AUC)	0.71
Decline risk	Precision / Recall	0.65 / 0.44
Decline risk	Brier score (calibration)	0.21
Volume forecast	Champion method	seasonal-naive
Volume forecast	Error (WAPE)	~72%

Read: the risk model reliably ranks who is most likely to drop; the volume forecast is directional, not exact, because individual small mills swing sharply quarter to quarter (that is what the high WAPE reflects). The dependable, actionable outputs are the risk bands and the sourcing shortlist.

4. Forecast — next quarter (FY2027 FQ2)

Across 510 scored suppliers, total forecast supply is about **50,800 MT**, or about **39,800 MT** after allowing for drop-risk. Largest expected volumes:

Supplier	Region	Usual/qtr (MT)	Forecast (MT)	Risk
Iniya Coir	Kangeyam	942	941	Low
Siva Selvi Fibers	Kangeyam	777	776	Low
Sughavesh Natural Fiber (HEC)	Salem	549	709	Low
Amman Coir	Kangeyam	708	708	Low
Palaniyanadvar Pith Block	Kangeyam	687	687	Low
Vetrivel Pith Block	Kangeyam	643	642	Low

Full per-supplier forecast is in the accompanying Excel workbook.

5. Which suppliers are declining

27 mills are at critical risk and 92 at moderate risk of a sharp drop. The highest-probability flags are small mills that stopped supplying last quarter:

Supplier	Region	Risk	Usual/qtr (MT)
Krishna Coir	Pollachi	82%	12
Kannamal Fibre	Pollachi	82%	6
PKS Coir	Pollachi	82%	5
J&D Pith Blocks	Pollachi	82%	6
AKS Coirs (Vivek)	Salem	80%	6
Karuppusamy Coir	Madurai	79%	4

More important commercially are the larger mills whose supply is weakening — these carry real volume and deserve a call or field visit first:

Supplier	Region	Usual/qtr (MT)	Last qtr (MT)	Risk
Shivanni Coirs	Pollachi	548	593	41%
Balu Coir	Pollachi	507	315	42%
Jai Sakthi Coir Products	Pollachi	397	423	40%
Venkateshwara Fibers	Pollachi	327	75	41%
Vishvajith Fiber	Pollachi	242	0	57%
Sri Krishna Coir Impexs	Pollachi	180	0	58%
Sabarigiri Fibres	Kangeyam	190	0	49%

6. Where to grow — sourcing opportunities

Big, mostly fast-growing mills where our share is low. Ranked by untapped volume:

Supplier	Region	Market (MT)	Our share	Untapped (MT)	Growth YoY
Siva Selvi Fibers	Kangeyam	3,410	16.8%	2,836	+34%
Vetrivel Pith Block	Kangeyam	2,819	2.7%	2,744	+246%
Iniya Coir	Kangeyam	3,830	32.3%	2,594	+42%
Balu Coir	Pollachi	2,501	2.0%	2,451	+131%
SP Exports (Sammy Coir)	Kangeyam	2,565	8.4%	2,350	+33%
Selvanayagiamman Fibers	Kangeyam	2,320	0.0%	2,320	+75%
Kadeswaran Pith Block	Kangeyam	2,294	7.8%	2,116	+13%

Kangeyam and Salem hold the most untapped, growing supply.

7. Supplier concentration (dependency)

Fiscal year	Our purchases (MT)	Top-5 share	HHI
FY2021	209,840	88.5%	0.20
FY2022	324,417	90.9%	0.45
FY2023	227,361	91.0%	0.34
FY2024	140,418	79.7%	0.30
FY2025	134,273	77.4%	0.27
FY2026	71,583	51.4%	0.11

Two clear signals: (a) our reliance on the top-5 suppliers has fallen sharply (91% to 51%) — good, we are less exposed; but (b) our total purchase volume has fallen about 78% from its FY2022 peak (324k to 72k MT). That volume decline is the single biggest question this report raises.

8. Regional picture (FY2026) and weather

Region	Market dispatch (MT)	Our offtake (MT)	Our share
Kangeyam	107,534	18,929	17.6%
Pollachi	77,229	11,123	14.4%
Madurai	32,049	5,171	16.1%
Salem	22,555	2,439	10.8%
Peravurani	3,791	400	10.6%

Kangeyam is both the largest market and our largest source; Salem is where our share is thinnest (10.8%). Weather: rainfall across all regions is at or below normal for the next 16 days, so production and drying should run freely near-term.

9. Implications and recommended actions

- **Grow:** pursue the Kangeyam and Salem opportunity list — the largest untapped, growing volume.
- **Protect:** field-check the larger weakening mills (Shivanni, Balu, Venkateshwara, Vishvajith, Sri Krishna Impexs) this week to confirm status and secure volume.
- **Investigate:** the ~78% decline in our purchase volume since FY2022 — the highest-impact strategic question.
- **Operate:** use the risk bands and opportunity ranking to drive weekly buying; treat exact forecast tonnage as guidance.

10. Reliability, caveats and data notes

- Forecasts are directional — small mills are volatile (WAPE ~72%). Risk bands and the shortlist are the dependable outputs.
- The MIR market panel undercounts (in some quarters our purchases exceed estimated market dispatch), so treat percentage shares as indicative.
- Data to request next for a big step-up: a stable supplier ID (to fix name matching), purchase price per MT, and mill capacity.
- Known raw-data items handled by the pipeline: future-dated field rows, a few dispatch-over-production inconsistencies, and extreme purchase entries (capped).

Accompanies: live dashboard (React), per-supplier forecast (Excel), weekly AI sourcing brief, speaker notes. Numbers in Metric Tonnes; refresh by re-running the pipeline.